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Industrial Hackathon

Problem Statements

1. AI-Based Material Test Certificate (MTC) Verification system

Background

Manufacturing and engineering organizations receive **Mill Test Certificates (MTCs)** from multiple suppliers and mills, often in different formats, layouts, and levels of clarity. Manual verification of these documents against applicable standards is time-consuming, error-prone, and heavily dependent on individual expertise.

Challenge

Design and develop an **AI-powered system** capable of automatically reading, interpreting, and validating **Material Test Certificates (MTCs)** against defined material standards or customer-specific requirements.

The solution should be able to:

- Extract key data such as:
  - ✓ Chemical composition
  - ✓ Mechanical properties
  - ✓ Heat / batch numbers
  - ✓ Test values and units
- Interpret data from **multiple document formats** (PDFs, scanned images, different layouts)
- Compare extracted values against:
  - ✓ International standards (e.g., ASME, ASTM, MIL) **or**
  - ✓ User-defined material specifications

Expected Outcome:

The proposed solution should:

- Flag compliance and non-compliance clearly
- Reduce manual verification effort and review time
- Minimize typographical and interpretation errors
- Create a **digital traceability and audit trail** for material validation

2. Energy Optimization of Industrial Air Compressors

Problem Statement

Compressed air systems account for nearly 10–30% of industrial energy consumption. Many systems operate at constant speed regardless of demand, leading to energy losses due to leakage, over-pressurization, and idle running.

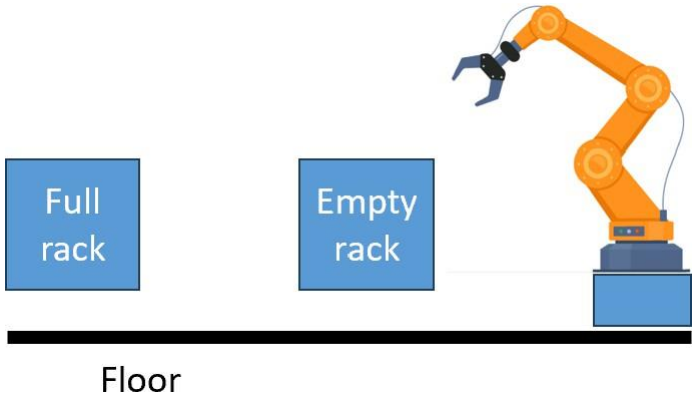
Design a smart energy optimization system that:

- Monitors pressure, load cycles, and leakages
- Adjusts compressor operation dynamically
- Reduces overall power consumption

Expected Outcome

- Working prototype/model (IoT-based or simulation-based)
- Real-time energy monitoring dashboard
- At least **10–20% projected energy savings**
- Cost-benefit analysis report
- Scalability plan for MSMEs

3. Rack changer requirement



Problem Statement

Sheet metal components are stored in racks from which robots pick parts for further production operations. Once a rack becomes empty, it must be replaced with a full rack to ensure uninterrupted production flow. Due to layout constraints, there is no access space available on the sides of the rack for replacement. Therefore, a mechanism must be developed to enable rack changing using only the available overhead (top) space safely and efficiently.

Deliverables

The project requires development of a feasible rack-changing mechanism that operates using overhead access. Conceptual layouts or sketches showing integration with the existing robotic setup must be provided. The proposal should clearly define the rack change sequence, safety considerations, and level of automation. Additionally, a feasibility assessment covering space requirements, load handling method, and estimated changeover time must be submitted.

4. Major rejection reduction - Sand defects in the top 3 rejection parts

Background

Irregularities in the final cast component that fail to meet design or quality standards, often due to sand mold degradation. These include sand inclusions (sand trapped in metal), sand washes (mold erosion), blowholes (trapped gas), swells (mold wall movement), and scabs (sand mold flaking). Most often caused by poor sand preparation (high moisture, low permeability), improper molding techniques, or high-temperature metal causing mold failure.

Challenge / Requirement

Sand defects are primary reasons for rejection of parts i.e, Hsg 933, Hsg Tur Brg 095, Lever Rocker 972, Provide a process-based solution to reduce the sand defects.

Deliverables

- Study of process parameters and actual process procedure
- POC demonstration

Known Bottlenecks / critical areas / Key factors to be addressed

- Cost benefits
- Customer expectations
- Able to handle quality inspection in Real time.

5. 2D Parts Identification Using On-the-Go AI Camera

Background

Manufacturing industries require accurate identification of 2D parts while they are moving on conveyor systems. Manual identification is time-consuming and prone to human error, leading to incorrect sorting, production delays, and quality issues.

Problem Statement

Design and develop an AI-based vision system capable of identifying and classifying 2D parts in real time while they move on conveyors, without interrupting production flow.

Expected Deliverables

- Camera + lighting setup proposal (industrial environment)
- Real-time part detection & classification model (with accuracy metrics)
- Handling of variations: orientation, speed, lighting, background
- Prototype demo (video/live) + simple UI/dashboard
- Integration approach for sorting/PLC trigger (optional/bonus)

6. 3D Parts Identification Using On-the-Go AI Camera

Background

In many production environments, parts with similar appearances but different shapes or dimensions must be accurately identified during movement. Traditional 2D vision is insufficient for detecting depth and dimensional variations.

Problem Statement

Develop an AI-powered 3D vision system that captures depth/geometry cues to identify and classify parts based on three-dimensional characteristics in real time.

Expected Deliverables

- 3D sensing choice justification (depth camera/structured light/stereo)
- 3D feature extraction or point-cloud based classification pipeline
- Accuracy and throughput validation plan
- Prototype demo + classification report
- Industrial feasibility notes (cost, robustness, maintainability)

7. Camera-Based Visual Inspection Along with Part Identification

Background

Manufacturing units often perform part identification and defect inspection as separate processes, increasing cycle time and operational costs. Manual inspection is inconsistent and may miss minor defects.

Problem Statement

Build an integrated AI camera system that simultaneously identifies parts and inspects them for visual defects, producing a combined decision (Part Type + Pass/Fail) in real time.

Expected Deliverables

- Combined pipeline: part ID + defect detection/segmentation
- Defect categories list and dataset/augmentation plan
- Quality decision logic with confidence scoring
- Reporting: defect snapshots + traceability log
- Prototype demo + performance report (precision/recall)

8. Multiple Items Identification at a Time

Background

In warehouses and production lines, multiple parts are frequently present in a single frame or batch. Manual counting and identification is inefficient and error-prone.

Problem Statement

Develop an AI system that detects, classifies, and counts multiple items in a single view with high accuracy and speed.

Expected Deliverables

- Multi-object detection model with per-class counts
- Support for overlaps/occlusions and cluttered trays
- Real-time counter + exportable count report
- Prototype demo + accuracy validation

9. AI-Based Material Counting

Background

Accurate material counting is essential for inventory management and production planning. Manual counting is labour-intensive, slow, and susceptible to errors.

Problem Statement

Create an automated AI-driven material counting system that reliably counts items in real time and can integrate with inventory/ERP systems.

Expected Deliverables

- Counting approach (detection, density estimation, tracking)
- Real-time counting UI + data export
- Inventory integration plan (API/CSV/DB)
- Accuracy, speed and error-handling report

10. Noise and Heat Reduction in Reciprocating Air Compressors

Problem Statement

Reciprocating compressors generate high noise and excessive heat, affecting workplace safety and equipment lifespan.

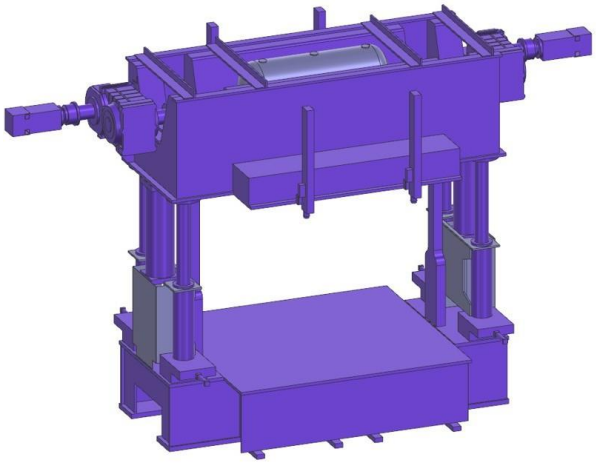
Develop a design modification or attachment that:

- Reduces noise levels
- Improves cooling efficiency
- Enhances operator safety

Expected Outcome

- Modified CAD design or prototype
- Thermal analysis or CFD simulation
- Measured decibel reduction comparison
- OSHA/industrial safety compliance study
- Implementation feasibility report

11. Die change requirement



Press



Die

Problem Statement

The production press requires periodic changing of hemming dies with dimensions 240 mm × 2400 mm and an approximate weight of 15 tons each. Currently, die change operations are time-consuming and may involve manual or semi-automatic handling methods, leading to extended downtime, safety risks, and dependency on operator skill. The requirement is to enable fully automated changeover of four different dies into and out of the press with minimal intervention. A robust, safe, and efficient automated die change system must be developed to reduce changeover time, improve safety, and enhance overall equipment productivity.

Die details:

- Hem Die size - 240mm X 2400mm
- Hem Die weight - 15 Tons

Deliverables

The project shall provide a fully automated die change concept suitable for handling 15-ton, 2400 mm long hemming dies. The proposal must include a detailed system concept covering die storage, transfer mechanism, alignment system, clamping interface, and integration with the press. A layout showing die staging for four dies, transfer path, and automation components (shuttle system, die lifters, bolster rollers, or rail-guided carts) must be submitted. The solution should include automation architecture, safety interlocks, estimated die change cycle time, required utilities, feasibility assessment, and a high- level Bill of Materials (BOM) with investment estimate.

12. Assembly Line Live Video Monitoring and Comparison with 3D Model

Background

Incorrect part placement, missing components, or improper assembly sequences cause defects and rework. Manual supervision may miss real-time assembly errors.

Problem Statement

Develop an AI-based live monitoring system that compares assembly video against a predefined 3D model/sequence, verifying correct positioning and order, and generates alerts on deviations.

Expected Deliverables

- Define reference model/sequence representation (3D model, CAD, checklist)
- Pose/part detection and matching method
- Deviation detection + real-time alerting
- Audit trail: timestamped deviations + evidence frames
- Prototype demo (live/recorded)

13. Vehicle Tracking System

Background:

Manufacturing organizations handle daily inbound and outbound vehicle movements for raw material pickup, supplier returns, and logistics operations. Limited real-time visibility of vehicle location, arrival status, and waiting times often leads to congestion at gates and stores, delays in material handling, and poor coordination between procurement, stores, and vendors.

Challenge:

Design and develop a digital vehicle tracking and monitoring system that provides real-time visibility of inbound and outbound logistics movements within a manufacturing facility.

The solution should be capable of:

- Tracking vehicle movement and location
- Monitoring arrival, departure, and completion of pickup/delivery tasks
- Identifying delays, idle waiting time, and bottlenecks
- Displaying logistics status through a clear, user-friendly dashboard

Expected Outcome:

The proposed solution should:

- Improve transparency across logistics operations
- Reduce vehicle waiting time at gates and stores
- Enhance coordination between procurement, stores, and vendors
- Support faster and more predictable material flow

14. Augmented Assembly Instructions (AR Assembly Guidance)

Background:

Manufacturing organizations often deal with complex, multi-step assembly processes, especially for customized products or new builds. Traditional assembly instructions such as drawings, SOPs, or manuals can be difficult to interpret on the shop floor, leading to longer learning curves, assembly mistakes, and inconsistent quality.

Challenge:

Design and develop an **Augmented Reality (AR)–based assembly instruction system** that provides **interactive, step-by-step visual guidance** to operators during complex assembly processes.

The solution should be capable of:

- Displaying assembly steps using AR overlays
- Highlighting:
- Correct part orientation and position
- Tool selection
- Torque values or critical parameters
- Assembly sequence and dependencies
- Supporting mistake-proofing by guiding the operator through the correct order of operations

Expected Outcome:

The proposed solution should:

- Reduce operator training and learning time
- Minimize assembly errors and rework
- Ensure consistency and repeatability in assembly quality
- Improve ease of understanding for complex or new assemblies

15. Low-Cost Portable Air Compressor for Rural Workshops

Problem Statement

Small-scale rural workshops (tyre shops, carpentry, welding units) require affordable and portable compressed air solutions. Existing compressors are expensive and power-intensive.

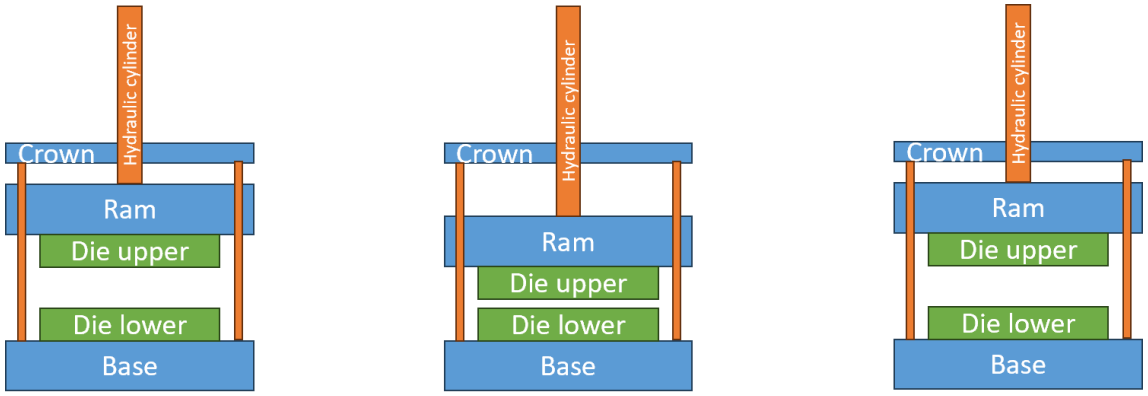
Design

- Low-cost
- Energy-efficient
- Portable air compressor
- Suitable for intermittent use in rural areas

Expected Outcome

- Compact mechanical design
- Cost under defined budget (e.g., ₹10,000 – ₹15,000 target)
- Power consumption analysis
- Noise reduction strategy
- Business model for rural deployment

16. Hydraulic press running cost reduction



Problem Statement

The hydraulic press currently consumes high energy during operation, particularly in the return stroke where significant power is required to lift the ram weight against gravity. During this upward movement, a considerable amount of energy is wasted, leading to increased operating and running costs. The existing system does not effectively recover or optimize the energy used during the return stroke. A solution is required to minimize energy loss during the return cycle through design modification, system optimization, or adoption of an improved hydraulic concept.

Deliverables

The project shall propose feasible technical solutions to reduce energy consumption during the return stroke of the hydraulic press. This may include design modifications, hydraulic circuit optimization, energy recovery systems, counterbalance arrangements, or new press concepts. A comparative analysis of current versus proposed energy consumption must be provided, along with estimated cost savings. The proposal should also include implementation feasibility, required modifications, investment estimate, and expected return on investment (ROI).

17. Re-Engineering the need for Nitriding in Mechanical Components:

Background:

Nitriding is widely used as a surface hardening process to improve wear resistance, fatigue life, and component durability. However, in many manufacturing applications, nitriding is applied as a standard practice without fully re-evaluating functional necessity, leading to increased cost, longer lead times, higher energy consumption, and environmental impact.

Challenge:

Design and propose alternative, optimized, or data-driven approaches to reduce, replace, or selectively apply nitriding while still meeting the required performance, reliability, and safety criteria of mechanical components.

The solution should be capable of:

- Functional requirements such as wear resistance, strength, and fatigue life
- Design optimization or geometry changes
- Material selection or material treatment alternatives
- Selective or localized surface treatment concepts
- Process simulation or analytical validation
- Cost, lead-time, and sustainability impact

Expected Outcome:

The proposed solution should:

- Reduce or eliminate unnecessary nitriding operations
- Lower manufacturing cost and cycle time
- Maintain or improve component performance and reliability
- Support sustainable and energy-efficient manufacturing

18. AI Based Solution for 3-Dimensional Visual Inspection

Background

Final products are being inspected manually for visual defects like aesthetics, markings, positions and letters printed.

Problem Statement

Develop an AI-based inspection solution (preferably with depth/3D assistance where needed) to detect visual defects and verify markings/print positions reliably.

Expected Deliverables

- Defect taxonomy + sample images plan
- Inspection pipeline (capture - preprocess - detect - decision)
- OCR/marking verification (where applicable)
- Prototype demo + inspection report/dashboard

19. AI-Based Predictive Maintenance System for Die Casting Machines

Background

Unexpected failures cause downtime, scrap and maintenance cost. Data is often available from sensors/PLC but not used for prediction.

Problem Statement

Build a predictive maintenance system that forecasts potential failures/abnormal conditions and recommends maintenance actions.

Expected Deliverables

- Data pipeline (sensor/PLC - storage - features)
- Failure/anomaly prediction model (classification/forecasting)
- Alerting + maintenance dashboard
- Evaluation plan (precision, lead time, false alarms)

20. Smart Tooling Management & On-Time Replacement System

Background

Tool wear and unexpected tool failure cause quality issues and downtime. Replacement is often time-based rather than condition-based.

Problem Statement

Develop a tooling lifecycle management system that predicts remaining useful life and triggers timely replacement alerts.

Expected Deliverables

- Tool usage tracking + digital tool history
- Wear/RUL prediction approach
- Replacement alert + planning dashboard
- Cost/benefit estimate (downtime avoided)

21. AI Vision System for Real-Time Foundry Process Monitoring

Background

Foundry processes are sensitive to variation; real-time monitoring can reduce defects early.

Problem Statement

Develop AI vision to monitor key process indicators and detect abnormal patterns that lead to casting defects.

Expected Deliverables

- Camera/thermal/vision setup concept
- Real-time detection of anomalies/defects
- Alerting + traceability log
- Prototype demo using sample videos/images

22. Energy Optimization in Manufacturing Using EMS & Analytics

Background

Energy usage is high and often poorly optimized across machines, shifts, and peak hours.

Problem Statement

Create an Energy Management System (EMS) analytics solution that monitors usage, identifies waste, and suggests optimization actions.

Expected Deliverables

- Energy data capture + KPI definitions
- Peak load and wastage analytics
- Optimization recommendations (load shifting, scheduling)
- Dashboard + reporting

23. Intelligent MRP Optimization & Demand Forecasting System

Background

Inaccurate forecasting leads to stockouts or excess inventory, affecting production and cost.

Problem Statement

Develop an ML-based forecasting + MRP optimization system to improve planning accuracy and inventory health.

Expected Deliverables

- Demand forecasting model
- MRP rules/constraints integration
- Inventory optimization metrics (service level, turns)
- Planning dashboard + export

24. AI-Based Automated Visual Inspection System

Background

Manual inspection is slow and inconsistent; automated inspection improves speed and repeatability.

Problem Statement

Build an AI-based automated inspection system for defect detection/classification with traceability.

Expected Deliverables

- Defect detection/classification model
- Pass/fail logic + confidence scoring
- Inspection log + evidence images
- Prototype demo + performance report

25. Smart Video Analytics for Shopfloor Safety & Productivity Monitoring

Background

Safety compliance and productivity monitoring is often manual and reactive.

Problem Statement

Develop smart video analytics to monitor safety (PPE, unsafe behavior) and productivity indicators (movement, idle time).

Expected Deliverables

- Safety analytics module + alerts
- Productivity insights (heatmaps, dwell time)
- Dashboards + reporting
- Privacy-aware design notes

26. AI Based Solution for Identifying and Segregating Crack Parts

Background

Micron-level cracks on small (≈5 mm dia) surfaces are being identified manually.

Problem Statement

Design a high-resolution AI vision solution to detect fine cracks and segregate defective parts automatically.

Expected Deliverables

- Optics proposal (macro lens, lighting, resolution requirement)
- Crack detection model (segmentation preferred)
- Measurement and confidence scoring
- Prototype demo + reject mechanism plan

27. AI Based Solution for Identifying and Segregating Defects in Pad Printing

Background

Pad-printed customer-specific information is inspected manually after curing when parts are stacked in trays.

Problem Statement

Build an AI system to verify printed information (correct content + legibility + alignment) and segregate defective parts.

Expected Deliverables

- OCR + template verification
- Print quality metrics (blur, contrast, completeness)
- Tray-wise batch inspection report
- Prototype demo + accuracy report

28. Automated Solution for Stacking & Counting the Parts

Background

Press components are accumulated in trays and counted/stacked manually.

Problem Statement

Develop an automated solution to count parts and validate stacking to reduce manual effort and errors.

Expected Deliverables

- Counting method with occlusion handling
- Stack height/arrangement validation (optional)
- Digital record per tray/batch
- Prototype demo

29. AI Based Voice-to-Text Shift Log with Storage & Communication

Background

Supervisors write shift information in a manual log book; handover between shifts can be inconsistent.

Problem Statement

Develop a voice-to-text system to capture shift notes, store them digitally, and communicate summarized handover to the next shift.

Expected Deliverables

- Speech-to-text + structured templates (Issue, Action, Pending)
- Searchable storage + shift-wise summary
- Notification/hand-over workflow (email/WhatsApp/in-app)
- Prototype demo

30. Automated Solution for Coating Components Based on Jig Movement

Background

Powder coating with auto gun + reciprocating setup is being done manually by operator.

Problem Statement

Design an automated control solution to trigger/optimize coating based on jig movement to ensure uniform coating and reduce wastage.

Expected Deliverables

- Motion/jig tracking approach (sensor/vision)
- Trigger logic + coating cycle optimization
- Uniformity validation plan
- Prototype/demo concept

31. AI Based Solution for Counting Boxes and Storing w.r.t Vehicle Number

Background

Boxes loaded in transporter vehicles at dispatch area: missing boxes observed.

Problem Statement

Develop a solution to count boxes during loading and map counts to vehicle number for traceability and mismatch alerts.

Expected Deliverables

- Box counting method (video/weight/RFID)
- Vehicle number capture (ANPR) and mapping
- Mismatch alert + dispatch report
- Prototype demo

32. AI Based Solution to Monitor and Optimize Man Movement

Background

Man movement to be monitored in shop floor to optimize movement.

Problem Statement

Develop video analytics to track movement patterns and provide optimization insights (heatmaps, bottlenecks).

Expected Deliverables

- Anonymized tracking approach
- Heatmaps + dwell-time analytics
- Optimization recommendations dashboard
- Prototype demo

33. AI Based Solution to Monitor and Auto Triggering for Safety Deviations

Background

Adherence to safety guidelines like pathway discipline, stacking height, PPE usage monitored manually.

Problem Statement

Develop AI monitoring that detects safety deviations and triggers alerts automatically.

Expected Deliverables

- PPE detection + rule engine
- Pathway/stacking violation detection
- Alerting + incident log
- Prototype demo

34. AI Based Solution to Measure Dimensions and Store in System

Background

Part dimensions are measured manually using Vernier, depth meter etc., based on samples.

Problem Statement

Develop a vision-based measurement system to measure dimensions and store results digitally with calibration.

Expected Deliverables

- Calibration procedure + measurement pipeline
- Dimension tolerance checking
- Digital storage/reporting
- Prototype demo

35. AI Based Solution to Capture Vehicle In-Out with Vehicle Number

Background

Vehicle in-out register maintained by security guard manually.

Problem Statement

Develop an ANPR-based automated entry/exit logging system with reports and traceability.

Expected Deliverables

- ANPR capture + timestamp logging
- Database + reporting dashboard
- Exception handling (unreadable plates)
- Prototype demo

36. Evaluating Additive Manufacturing for Disc/Trim Components:

Background:

Disc and trim components used in mechanical and valve assemblies are often manufactured through conventional machining processes. While machining ensures accuracy, it can result in high material wastage, long lead times, and increased cost—especially for low-volume or complex geometries. Additive manufacturing offers an alternative approach but requires careful evaluation of design feasibility, performance, and cost.

Challenge:

Design a **decision framework and component redesign approach** to determine when **disc or trim parts can be manufactured using additive manufacturing (3D printing) instead of traditional machining**, without compromising functional requirements or reliability.

The solution should consider:

- Functional and performance requirements of disc/trim parts
- Redesign for Additive Manufacturing (DfAM) principles
- Material selection suitable for additive processes
- Strength, wear, and dimensional accuracy considerations
- Cost, lead time, and material utilization comparison
- Post-processing or hybrid manufacturing needs

Expected Outcome:

The proposed solution should:

- Identify suitable disc or trim components for additive manufacturing
- Demonstrate potential cost and lead-time reduction
- Ensure functional equivalence or improvement over machined parts
- Provide a clear recommendation framework for manufacturing selection

37. Predictive Maintenance System for Air Compressors

Problem Statement

Unexpected compressor failures cause production downtime and financial losses. Current maintenance practices are mostly preventive or reactive.

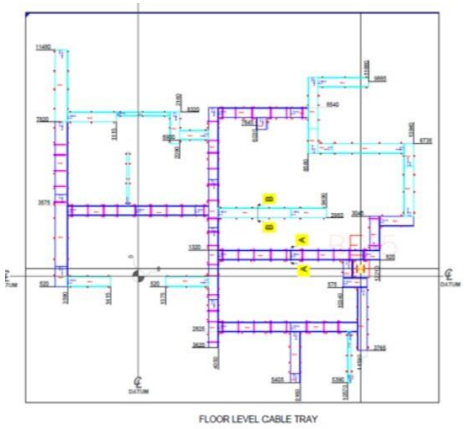
Develop a predictive maintenance solution using:

- Vibration sensors
- Temperature monitoring
- Pressure fluctuation tracking
- AI/ML failure prediction models

Expected Outcome

- Sensor-integrated monitoring system
- Failure prediction algorithm
- Early fault detection (bearing failure, valve leakage, overheating)
- Maintenance alert mobile/web interface
- Reduced downtime estimation report

38. Floor marking requirement







Problem Statement

During installation of machines, tools, and cable trays, floor marking is currently performed manually by transferring dimensions from AutoCAD layout drawings onto the shop floor. After marking reference lines and equipment positions, installation activities are carried out based on these markings. This manual process is time-consuming and highly dependent on operator accuracy. As a result, marking errors can occur, leading to misalignment, rework, and installation delays.

Deliverables

The objective is to identify and propose modern technologies or equipment that can automate or improve the accuracy of floor marking. The solution should reduce marking time, minimize human error, and improve installation precision. A comparative assessment of available technologies (e.g., laser projection systems, total stations, AR-based layout tools, or robotic marking systems) should be included. The proposal must also outline implementation requirements, expected accuracy, cost considerations, and integration with existing AutoCAD or digital layout files.

39. Smart Multi-Compressor Load Sharing System

Problem Statement

In large plants, multiple compressors operate simultaneously without optimized load sharing, leading to uneven wear and inefficiency.

Design an intelligent load-sharing control system that:

- Automatically sequences compressors
- Balances runtime hours
- Maintains optimal pressure
- Minimizes energy consumption

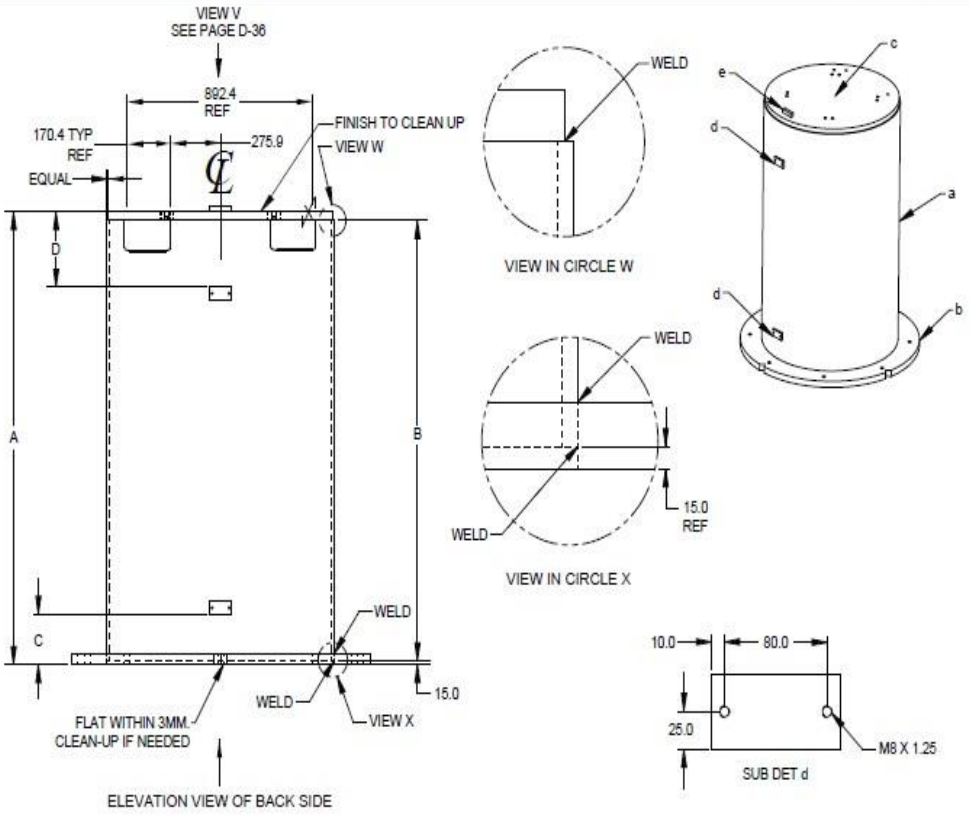
Expected Outcome

- PLC/Arduino/IoT-based controller model
- Pressure stabilization demonstration
- Equalized runtime distribution analysis
- Simulation results comparing conventional vs smart system
- Industrial implementation roadmap

40. Robot Riser - Welding Automation

ROBOT RISER
R2000iC 165F/210F/125L

GLOBAL STANDARD COMPONENTS
NAAMS™
Assembly 06/21/24



FEA WAS DONE WITH MAX LOADING: Mv 73.50 kNm

THE CONCRETE USED IN THE FEA AND HILTI PROFIS WAS A MINIMUM OF 6" DEEP SINGLE SLAB WITH A COMPRESSIVE STRENGTH OF 4000 PSI WITH 3/4" DIAMETER HILTI HIT-Z AND HAS-E ANCHORS.

NAAMS CODE	A ±5.0	B	C	D	WEIGHT
ARR820	300	245.0	N/A	N/A	567 kg
ARR830	600	545.0	300MM	N/A	667 kg
ARR840	900	845.0	450MM	N/A	765 kg
ARR850	1200	1145.0	200MM	300MM	864 kg
ARR860	1500	1445.0	200MM	300MM	963 kg
ARR870	1800	1745.0	200MM	300MM	1062 kg

A

SEE DRAWING ON NEXT PAGE

INCH STOCK:
(a) 1 ~ 1065MM DIA X 1/2" W/T x B - ASTM A36
(b) 1 ~ 1 3/4" X 1400MM DIA - ASTM A36
(c) 1 ~ 1 3/4" X 1050MM DIA - ASTM A36 MADE FROM HOLE IN SUB DET "b"
(d) 0, 1 or 2 ~ 1/2" X 2" X 100MM - ASTM A36 SEE CHART
(e) 1 ~ 1" X 1" X 50MM - ASTM A36

© 2024 USCAR

Riser Details: Refer www.naamsstandards.org (Pg:35-37)

Problem Statement

The Robot Riser assembly requires MIG welding at three designated weld locations as part of its manufacturing process. Currently, the welding process is manual or semi-automatic, which may result in inconsistent weld quality, higher cycle time, and dependency on operator skill. Considering the riser is a structural component with significant load-bearing requirements, uniform weld penetration, dimensional stability, and repeatability are critical. Therefore, an automated welding solution is required to improve weld consistency, productivity, process control, and overall manufacturing efficiency.

Inputs:

- Welding Type = MIG welding
- Areas to be welded = 3

Deliverables

The project shall provide a basic concept for an automated MIG welding solution suitable for welding the three identified areas of the Robot Riser. A proposed system layout including robot, positioner, and welding equipment integration must be submitted. A detailed concept-level Bill of Materials (BOM) covering robotic system, welding equipment, fixtures, safety systems, and auxiliary components shall be included. Additionally, the proposal must outline the welding sequence, expected cycle time, technical feasibility, and key benefits compared to the existing manual process.

41. Rework reduction in the top 3 repeatedly reworked parts

Background

An part in manufacturing cycle undergoes series of operations such as Metal forging,, casting, stamping, and sheet metal fabrication etc., some of the parts may proceeded to final fabrication skipping the series of operations which leads to rework. Rework in manufacturing significantly affects the time line and production.

Challenge / Requirement

Provide an end-to-end mechanism for Cognitive Quality Inspection in Manufacturing operations.

Deliverables

A process-based solution for Quality Inspection on a top 3 rejection parts to reduce reworks in 3 repeatedly rework parts.

Known Bottlenecks / critical areas / Key factors to be addressed

- Cost benefits
- Customer expectations
- Able to handle quality inspection in Real time.